Course Work Specification:

Academic term - Autumn 2025/2026

CC7183 Data Analysis and Visualisation

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Content

Introduction to Coursework

The aim of this coursework is to empower you to transform raw data into actionable knowledge. You will prepare, analyse, and interpret a real-world dataset, implement a chosen analytical model, and communicate your findings both through a professional academic report and a dynamic Power BI dashboard. This project mirrors the challenges faced by data analysts and scientists in real organisations, where extracting insight from complex and imperfect data is a key skill.

This coursework is designed not only to assess your technical expertise but also to develop your ability to think critically, make evidence-based arguments, and present results in ways that influence decision-making.

By working with authentic datasets, you will experience the realities of modern data science: handling missing or noisy data, justifying methodological choices, and translating quantitative outputs into meaningful conclusions.

The aim of this coursework is to prepare data for analysis, implement, a model and tell a story by visualising the data within the chosen scenario. This is an individual coursework. The coursework contributes ***60% to the overall CC7183 module mark.***

Scenario

You are required to independently select a topic of personal or professional interest, obtain an appropriate dataset (government or publicly available, but not Kaggle), and apply one of the analytical models listed below. Your work must showcase both technical depth and storytelling ability. The topic should allow you to explore meaningful questions and generate actionable insights. You are encouraged to think critically about the data, uncover patterns, and provide interpretations that go beyond descriptive statistics.

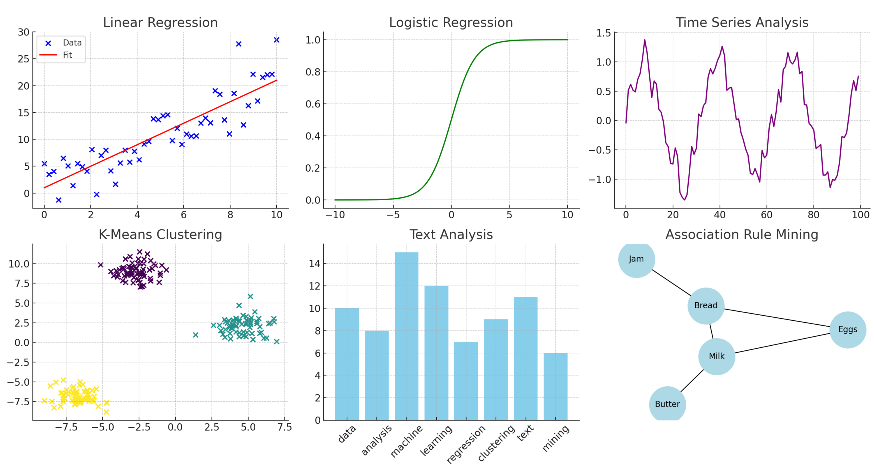
The final deliverables should highlight:

- Effective data preparation and cleaning.
- Appropriate application of your chosen analytical model.
- Clear and impactful data visualisations.
- Critical interpretation and reflection on findings.

Selection of models

Choose one of the following models as the foundation of your project:

1. Linear Regression
2. Logistic Regression
3. Time Series Analysis
4. K-Means Clustering
5. Text Analysis
6. Association Rule Mining



Example Project Topics

- Trends in air quality and their impact on urban health.
- Crime and safety analysis using police open data.
- Gender representation in STEM education and employment.
- Understanding patterns in food security and agricultural output.
- Public transport demand forecasting using time series.
- Exploring global economic indicators and development outcomes.

Suggested Data Sources

Data Source	Link	Subject Area
NYC Open Data	https://opendata.cityofnewyork.us/data/	City Data, Government
Office for National Statistics (ONS)	https://www.ons.gov.uk/	Economy, Population, Social
UCI Machine Learning Repository	https://archive.ics.uci.edu/	Machine Learning, Healthcare, Finance
Dryad	https://datadryad.org/search	Environment, Agriculture, Forestry, Medical and Health
UN Data	http://data.un.org/Explorer.aspx	Global, Health, Economics
UK Government Data	https://www.data.gov.uk/	Government, Health, Agriculture, etc.
US Government Data	https://catalog.data.gov/dataset	Government, Health, Agriculture, etc.
World Bank Data	https://data.worldbank.org/	Economics, Global Development
London Datastore	https://data.london.gov.uk/dataset/	Demographics, Environment, Planning, Health, Education, etc.
Google Dataset Search	https://datasetsearch.research.google.com/	Multi-disciplinary
EU European Data	https://data.europa.eu/en	Government, Economy, Environment
UK Police Data	https://data.police.uk/docs/	Crime, Safety, Law Enforcement
Open Power System Data	https://data.open-power-system-data.org/	Energy, Power, Renewable Energy
FAO – Food and Agriculture Organization	https://www.fao.org/faostat/en/#data	Agriculture, Food Security, Fisheries
Humanitarian Data Exchange (HDX)	https://data.humdata.org/dataset?	Humanitarian, Crisis Response, Refugees

Note: Use of Kaggle datasets is not permitted. You may utilize any of the sources listed above or other open-access online datasets, provided they originate from government sources or other reputable sites.

Marking Scheme

1. **Proposal Submission (5 marks):** Proposal for the journal-styled report is due in week 4, which should include the selection of data and a suitable model to visualise the data in Power BI.

2. **Peer Feedback / Continuous Development (15 marks):** Students need to show their progress by week 6 and week 7 and obtain feedback from peers to guide further development.

3. **Academic Journal (55 marks):**

   (a) At the end of the analysis, students are required to write an academic journal in IEEE referencing style. The journal should present the analysis in an academic manner.

   (b) The journal should include an introduction, the model(s) used – literature review and methodology, findings, and a critical reflection with recommendations.

   (c) The journal must be submitted by week 12.

4. **Dashboard Submission (25 marks):** Students will submit their Power BI Dashboard by week 12. The dashboard should clearly present the analysis and findings in a professional and visually effective manner.

Assessment Details		Key Dates	
Component	**Marks**	**Deadline Week**	**Due Dates**
Project Proposal	5 marks	Week-4	24th Oct 2025
Peer Review (Submission &...	15 marks	Week-6 & 7	7th Nov 2025, 14th Nov 2025
Academic Journal Report	55 marks	Week-12	19th Dec 2025
PowerBI Dashboard	25 marks	Week-12	19th Dec 2025

Figure 1: Deadlines for coursework

Examples of Data Analysis Models

Model Name	Type	Example Data
Linear Regression Model	Dependence of variables	**Example 1:** Predicting house prices based on features like square footage, number of bedrooms, and neighbourhood quality. **Example 2:** Estimating the amount of electricity consumption based on temperature, time of year, and household size.
Time Series Analysis	Trend over long term	**Example 1:** Forecasting monthly sales for a retail store based on historical sales data. **Example 2:** Predicting future stock prices based on past stock prices and trading volume.
Logistic Regression Model	Binary Classification	**Example 1:** Predicting whether a customer will churn (yes/no) based on their usage patterns, demographics, and customer service interactions. **Example 2:** Classifying emails as spam or not spam based on the presence of certain keywords and other metadata.

Model Name	Type	Example Data
K-Means Clustering	Clustering / grouping data	**Example 1:** Segmenting customers into distinct groups based on purchasing behaviour for targeted marketing. **Example 2:** Grouping similar documents together for topic modelling in text mining.
Text and Graph Analysis	Analysing unstructured data	**Example 1:** Sentiment analysis of social media posts to determine public opinion on a product. **Example 2:** Analysing relationships in a social network to identify influential users and community structures.
Association Rule	Identify relationships / patterns in large datasets	**Example 1:** Market basket analysis to find associations between items frequently bought together, such as milk and bread. **Example 2:** Identifying patterns in website clickstream data to understand common navigation paths taken by users.

Submission Guidelines

All submissions must be uploaded to the Virtual Learning Environment (VLE) by 15:00 on the deadline date.

- The report should follow IEEE referencing style and be 2500 words with +/-10%.
- The Power BI dashboard should be created on PowerBI and uploaded on a separate folder.
- The final dashboard must be included in the appendix and also submitted separately on weblearn.

Academic Integrity

Work must comply with the University's academic integrity policy. Proper referencing of all datasets, code, and literature is mandatory. Plagiarism or fabrication will result in penalties.

At London Metropolitan University, plagiarism is treated as a serious academic offence. If plagiarism is detected, the maximum mark that can be awarded for the assessment is capped at 40%. In cases of severe or repeated plagiarism, a mark of 0 will be given.

All submissions will be subject to plagiarism testing, including the detection of unreferenced use of words or material taken from the internet, websites, journals, and books. Results of the plagiarism check will be recorded on Weblearn.